

The total energy of a simple harmonic oscillator for displacement and velocity  $x, v$  in the system.

$$E = \frac{1}{2}mv^2 + \frac{1}{2}kx^2 \text{ which is conserved for any respective } x, v$$

Divide by  $E$  and rewrite,

$$\frac{mv^2}{2E} + \frac{kx^2}{2E} = 1$$

$$\frac{v^2}{\left(\frac{2E}{m}\right)} + \frac{x^2}{\left(\frac{2E}{k}\right)} = 1$$

Suppose we have a plot with  $v$  on  $y$ -axis and  $x$  on  $x$ -axis. Compare with the ellipse equation  $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$  where  $a$  is the semi-major axis and  $b$  is the semi-minor axis.

Then the  $x-v$  equation shows an ellipse with  $\sqrt{\frac{2E}{k}}$  as the semi-major axis and

$\sqrt{\frac{2E}{m}}$  as the semi-minor axis.

Therefore, conservation of energy in the simple harmonic oscillator system leads to an ellipse in the  $(x, v)$  space.